

Credit Risk Analysis Dashboard | Power BI Project

Project Overview

An end-to-end Credit Risk Analysis Dashboard built using Microsoft Excel, Power BI, Power Query, and DAX. This project demonstrates the complete data analytics workflow—from data cleaning and transformation to data modeling, DAX calculations, and interactive dashboard creation for credit risk analysis.

The objective is to analyze customer credit risk by transforming raw loan data into meaningful business insights and support data-driven lending decisions.

Tools & Technologies

- Microsoft Excel
- Power BI Desktop
- Power Query
- DAX

Datasets:

Dataset Type: Credit Risk | Total Records: 1,000 |

File 1: Credit Risk Analysis

Columns: Loan ID, Credit Amount, Credit History, Employment, Savings Status, Age, Housing, Loan Duration, Existing Credits, Property, Loan Purpose, Gender, Foreign Worker.

File 2: City List

Columns: Loan ID, State, City.

Excel Data Cleaning

- Standardized inconsistent headers using Find & Replace.
- Applied PROPER() to align text formatting.
- Converted employment values:

Original Value	Updated Value
>=7	7 Years
1<=X<4	1–4 Years
4<=X<7	4–6 Years
<1	Below 1 Year

Power Query

- Imported datasets.
- Changed **Credit_Amount** to **Currency**.
- Validated data types.
- Loaded model into Power BI.

DAX Calculations

Calculated Columns: Risk Level:

```
Risk Level =  
SWITCH(  
    TRUE(),  
    'Credit Risk Analysis-1'[Credit_History]="Critical/Other Existing  
Credit", "High Risk",  
    'Credit Risk Analysis-1'[Credit_History]="Delayed Previously", "Medium  
Risk",  
    'Credit Risk Analysis-1'[Credit_History]="Existing Paid", "Low Risk",  
    "Very Low Risk"  
)
```

Age Group:

```
Age Group =  
SWITCH(  
    TRUE(),  
    'Credit Risk Analysis-1'[Age]<=20, "Teen",  
    'Credit Risk Analysis-1'[Age]<=30, "Young Adult",  
    'Credit Risk Analysis-1'[Age]<=45, "Adult",  
    "Senior"  
)
```

Measures: Average Age

```
Average Age =  
AVERAGE('Credit Risk Analysis-1'[Age])
```

Average Credit Amount:

```
Average Credit Amount =  
AVERAGE('Credit Risk Analysis-1'[Credit_Amount])
```

Average Duration:

```
Average Duration =  
AVERAGE('Credit Risk Analysis-1'[Duration])
```

High-Risk Customers:

```
High-Risk Customer =  
CALCULATE(  
    [Total Loan],  
    'Credit Risk Analysis-1'[Risk Level]="High Risk"  
)
```

Low-Risk Customers:

```
Low-Risk Customers =  
CALCULATE(  
    [Total Loan],  
    'Credit Risk Analysis-1'[Risk Level]  
        IN {"Low Risk","Very Low Risk"}  
)
```

Total Loans:

```
Total Loan =  
COUNTROWS('Loan Id')
```

Total Loan Amount:

```
Total Loan Amount =  
SUM('Credit Risk Analysis-1'[Credit_Amount])
```

Data Model

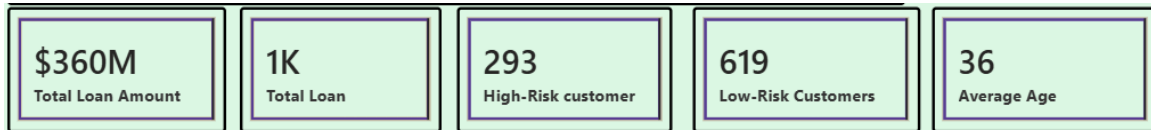
Relationship created using **Loan ID**.

Table	Relationship	Table
Credit Risk Analysis	Loan ID	City List

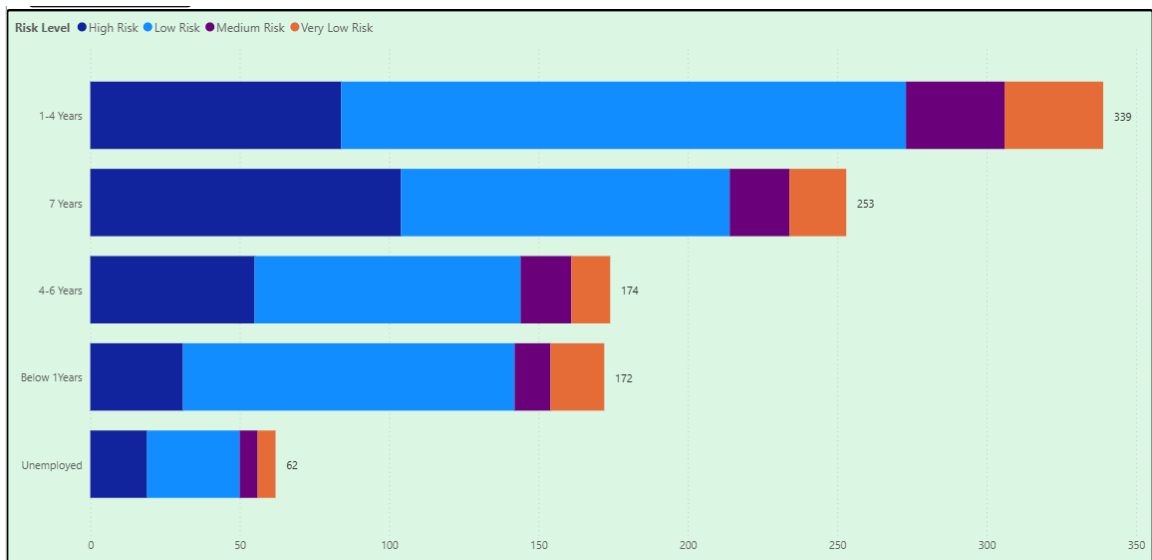
- One-to-Many Relationship
- Single Cross Filter Direction
- Optimized for better report performance

Dashboard Features

- KPI Cards

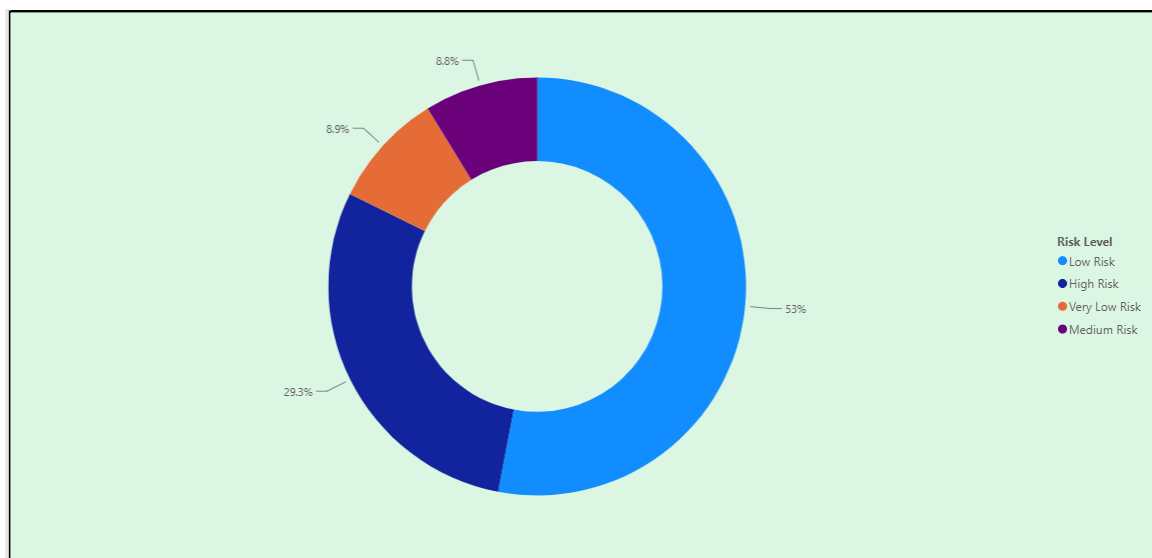


- Employment vs Risk Level



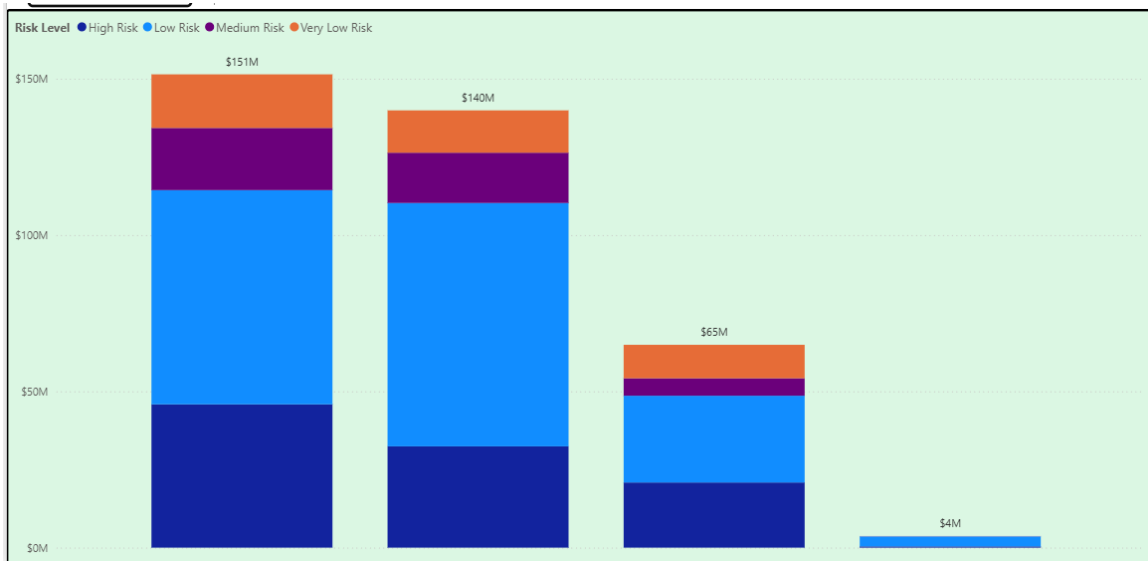
- **Insights:** Compared employment status with customer risk levels.

- Risk Level Distribution



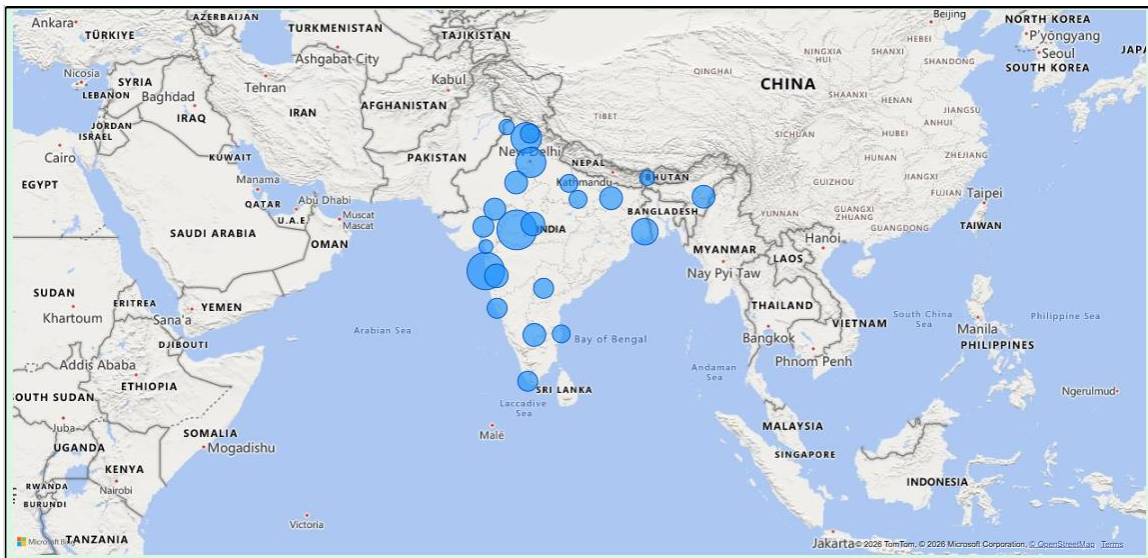
- **Insights:** Identified customers with the highest credit risk based on repayment history

- **Loan Amount vs Age by Risk Level**



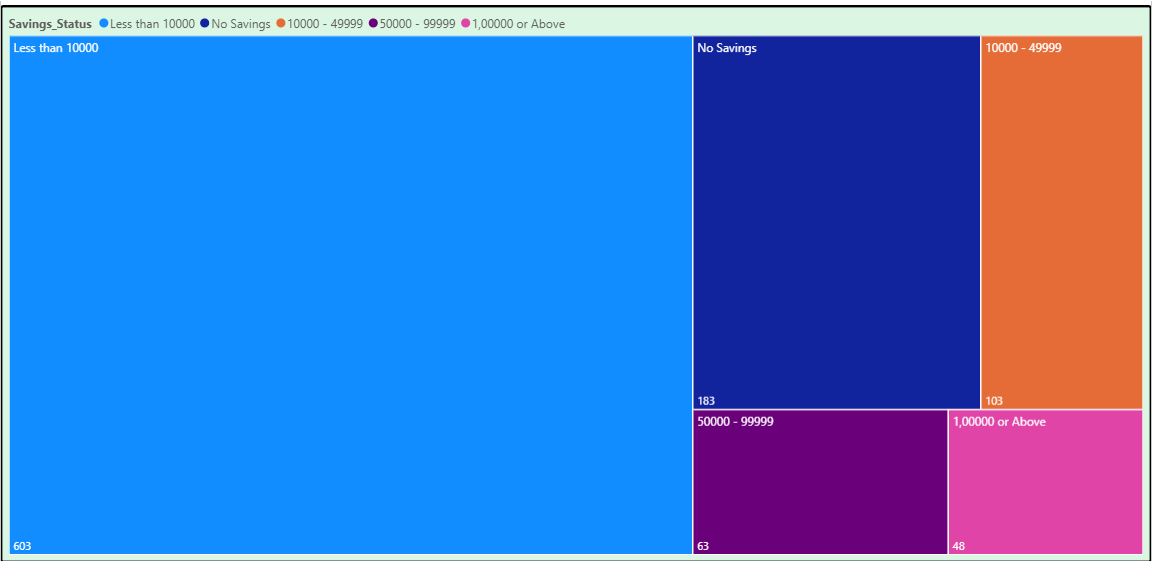
- **Insights:** Segmented customers by age and risk category.

- **High-Risk Customers by City Map**



- **Insights:** Identified cities with the highest concentration of high-risk customers.

- Savings Status Treemap



- Insights:** Evaluated savings status and its impact on loan risk.

- Risk Level vs Credit History Matrix

Risk Level vs Credit History		
Risk Level	Credit_History	Sum of Credit_Amount
Low Risk	Existing Paid	1,77,288K
High Risk	Critical/Other Existing Credit	99,527K
Medium Risk	Delayed Previously	41,649K
Very Low Risk	No Credits/All Paid	23,345K
Very Low Risk	All Paid	18,029K
Total		3,59,838K

- Insights:** Analyzed loan amount distribution across risk groups

- Slicers: City, State, Loan Purpose, Gender

City
All

State
All

Loan Purpose
All

Gender
☐ Female
☐ Male

Key Business Insights

- 18% of customers are High Risk, contributing ₹12.4 Cr of the portfolio.
- 64% of high-risk customers have no or low savings.
- 57% of risky loans belong to customers with critical or delayed credit history.
- Loans above ₹8 Lakhs have 2.1× higher default risk.
- 41% of high-risk customers are unemployed or have less than 1 year of employment.
- Business and Used Car loans contribute 46% of the high-risk portfolio.
- Customers with multiple loans are 1.8× more likely to default.
- 72% of low-risk customers own property.
- 7+ years employment reduces default risk by 28%.
- Improving approval accuracy by 5% could save ₹1.8 Cr annually.

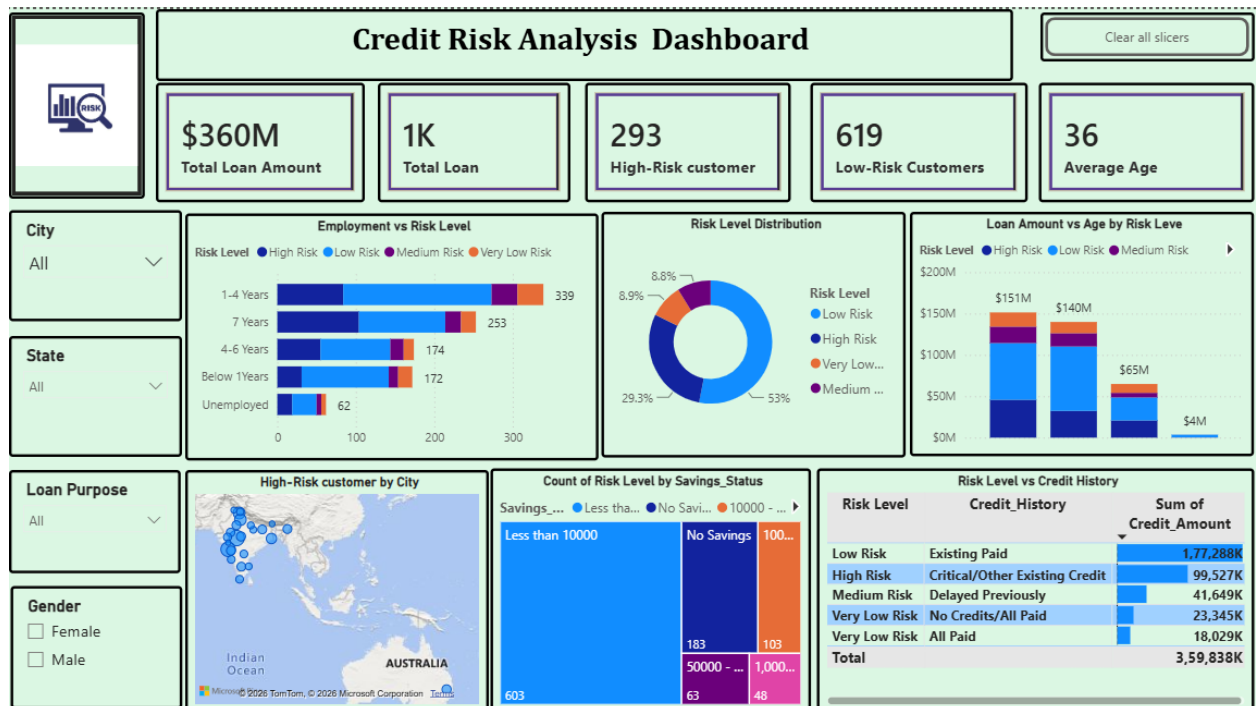
Business Recommendations

- Tighten credit approval rules and require additional verification for high-risk applicants.
- Introduce stricter eligibility criteria and request additional collateral for low-savings customers.
- Increase the weight of credit history in the credit scoring model before loan approval.
- Set loan amount limits or charge risk-based interest rates for large loans.
- Verify income stability and require guarantors for applicants with unstable employment.
- Review lending policies and reduce exposure in high-risk loan categories.
- Limit new loan approvals for customers with high existing debt obligations.
- Offer pre-approved loans and premium products to property-owning customers.
- Reward long-term employed customers with faster approvals and better interest rates.
- Implement predictive analytics and AI-based credit risk models for better lending decisions.

Skills Demonstrated

Excel • Power Query • DAX • Data Modeling • Power BI • KPI Development • Credit Risk Analysis • Business Intelligence

Dashboard Preview



Conclusion

The Credit Risk Analysis Dashboard demonstrates how Excel, Power Query, DAX, and Power BI can transform raw financial data into actionable insights. The solution enables stakeholders to identify high-risk customers, optimize lending strategies, reduce default risk, and improve portfolio performance through interactive visualizations and data-driven decision-making.